

Date: July 2, 2026

Subject: Submission of original research article: “*Structure-Guided Prioritization of Divergent Virulence and Resistance Candidates in an Open Pangenome Clinical Escherichia coli Isolate*”

Dear Editor-in-Chief,

We are pleased to submit our original research manuscript entitled, “**Structure-Guided Prioritization of Divergent Virulence and Resistance Candidates in an Open Pangenome Clinical *Escherichia coli* Isolate,**” for consideration as a research article in *Current Microbiology*.

Determining the biological function of hypothetical proteins is a major bottleneck in microbial genomics. This is particularly true for the accessory genome “dark matter” of clinical pathogens, which frequently conceals novel virulence factors and antimicrobial resistance (AMR) determinants that escape traditional sequence-alignment search tools. To address this bottleneck, our study introduces an integrated, sequence-independent **pangenome-to-structure computational discovery pipeline**.

By combining pangenomic singleton filtering with high-throughput 3D structural modeling (ESMFold/AlphaFold3), Foldseek fold homology matching, protein language model (PLM) embeddings (ESM-2), and explicit-solvent molecular dynamics (MD) simulations, we identified and computationally prioritized four high-priority candidate singletons in the genome of a clinical multidrug-resistant (MDR) extraintestinal pathogenic *Escherichia coli* (ExPEC) ST354 isolate (QA5221) from Algeria:

- **GNAT_KA27** (KNGPFPPJ_02769): A GCN5-related N-acetyltransferase (GNAT) fold with active-site docking supporting kanamycin binding capacity and dynamic stability over a 127.78-ns explicit-solvent MD trajectory.
- **Ehly_61** (KNGPFPPJ_00061): A pore-forming enterohemolysin-like cytolysin associated with lysogenic conversion, analyzed via 100-ns MD simulations showing membrane-free hydrophobic collapse.
- **OAgP_161** (KNGPFPPJ_03161): A highly divergent O-antigen polymerase (Wzy) embedded in the lipopolysaccharide (LPS) gene cluster, indicating serotype switching.
- **OAT_371** (KNGPFPPJ_04371): A plasmid-associated outer-membrane autotransporter candidate representing a divergent Type V secretion system.

All four candidates are ultra-rare singletons showing depressed GC/GC3 content consistent with foreign horizontal acquisition.

We believe this study is a perfect fit for the scope of *Current Microbiology*, as it combines cutting-edge computational biophysics and machine learning with clinical bacteriology, offering a generalizable methodology for uncovering hidden functional diversity in pathogenic bacterial pangenomes. Due to the multi-layered analysis across four distinct candidates, our manuscript (8,043 words, 8 figures) slightly exceeds the standard word and figure limits. We kindly request your consideration for an exception in this case.

We confirm that this manuscript represents original work and has not been published previously, nor is it currently under consideration for publication elsewhere. All authors have read and approved the final manuscript for submission, and we declare no conflicts of interest.

Thank you very much for your time and consideration of our work. We look forward to hearing from your editorial team.

Sincerely,

Sarra Benmoumou-Hosni

On behalf of all co-authors: Atika Meklat and Ikram Haleche

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